

Highlights

Fast Time Series Anomaly Detection Using An Efficient Periodicity-Aware Neural Network

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- Introducing an Efficient Periodicity-Aware Anomaly Detection Network for edge devices on production lines.
- Proposing a Sine-Cosine Transform Module to introduce periodic inductive bias for time-series anomaly detection.
- Performing extensive experiments on six public data sets with 14 baselines, which provide benchmarks for the community.

Fast Time Series Anomaly Detection Using An Efficient Periodicity-Aware Neural Network^{*}

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ABSTRACT

Nowadays, Anomaly Detection (AD) in time series plays important roles in many fields, such as industrial production, finance, network security and medical health. Recently, the advancement of Internet of Things (IoT) has led to the emergence of intelligent manufacturing and the widespread usage of edge devices. As a result, there is an increasing demand for rapid and effective anomaly detection in the time series data captured at the edge. However, it is still challenging to design an efficient and effective time series anomaly detection approach, which is suitable for edge environments, due to real-time constraints and limited resources. To address this issue, we propose a novel anomaly detection neural network, namely, Efficient Periodicity-Aware Anomaly Detection Network (EPAAD-Net). This network is mainly designed on top of one-dimensional convolutions, which ensure low inference latency for edge deployment. We also deliberately design a Sine-Cosine Transform Module (SCTM), which projects the extracted features onto periodic basis functions to introduce a periodic inductive bias, thereby boosting the performance of the network. To capture the long-term dependencies between data points, we further incorporate a cross-attention module into the EPAAD-Net. Experimental results demonstrate that the proposed EPAAD-Net achieves competitive detection accuracy compared to 14 baselines across six data sets, while offering higher inference efficiency. We believe that these promising results should be due to the joint use of the simple network architecture and the periodicity-aware SCTM.

1. Introduction

With the increasing prevalence of edge devices in industrial production, modern industrial manufacturing lines continuously generate a substantial volume of time series data, which are normally collected using different edge sensors or IoT devices [1]. Since these data depict temporal variations in various production factors, such as the equipment status and environmental conditions, they are important to monitoring the status of devices and analyzing the production process [2]. As the dimensionality of the data continues to expand, however, the complexity of these tasks is also escalated. In this situation, it is a formidable challenge to achieve timely and effective Anomaly Detection (AD) in the complex high-dimensional data [3, 4]. Fast and effective time series anomaly detection becomes imperative and particularly challenging to industrial production [5].

Given a data set, anomaly detection aims to discover data points which deviate from the anticipated patterns or significantly differ from the majority of the data set [6]. Basically, these data points present irregular occurrences or exceptional circumstances. The traditional anomaly detection methods mainly include statistical approaches [7, 8],

rule-based techniques [9, 10] and machine learning-based methods [11, 12, 13, 14, 15]. Since statistical approaches often use Principal Component Analysis (PCA) [7] or Markov chains [8] to model data, they usually encounter limitations in processing the intricate time series data even if they are straightforward and interpretable. In addition, these approaches present less effectiveness when dealing with the nonlinear data and are susceptible to the nuances in data.

Regarding the rule-based anomaly detection methods, they usually rely on a set of hand-crafted rules in order to identify the anomalies in data [9, 10]. These rules can be obtained based on the knowledge and experience of domain experts. The rule-based methods typically exhibit the high flexibility because the rules are adaptable to real-world scenarios and they are thus applicable to diverse contexts. Although these methods do not require a large set of training data and the results produced are readily interpretable, they still encounters three challenges. First, the rules that those methods use are normally formulated based on the known knowledge and experience, which prevent them from being popularized to unknown anomalies or scenarios. Second, the formulation of those rules may be influenced by subjective factors, which impairs the reliability of the rule-based methods. Third, the formulation process requires a significant amount of time and manpower costs, particularly, when the complexity of data is heavy, which limits the application of these methods.

In contrast, the machine learning-based anomaly detection methods typically build a model by learning patterns from the training data for anomaly detection, which normally involve regression models [11], Support Vector Machines

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^{*}Code and models will be made available at <https://github.com/INDTLab/EPAAD-Net> upon the acceptance of the paper.

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(SVMs) [12, 14, 13] and K-means clustering [15]. However, the model trained using these algorithms may encounter a challenge in interpreting the results that they predict and may also be overfitted by the training data. As a result, the generalization ability of the model is diminished.

In terms of the annotations of the training data, machine learning approaches can be categorized into three classes, including supervised learning [6], unsupervised learning [16] and weakly supervised learning [17]. For the supervised learning methods, the model is trained using the input data along with the corresponding labels. Benefiting from the label data, these methods typically achieve the better performance and are suitable for the scenarios with the abundant labeled data. In the case that the labeled data is limited, weakly supervised learning approaches can be used. However, the time series data usually lack labels in the real-world industrial scenarios because the data annotation operation on a large amount of time series data is time-consuming and expensive. In this context, unsupervised learning methods are favored for anomaly detection in time series data [15, 14, 7].

Although many traditional time series anomaly detection methods have been developed on top of unsupervised learning techniques, they often overlook temporal correlations and tend to disregard inter-variable dependencies when handling the multivariate time series data, leading to the suboptimal results. With the advancement of deep learning techniques [18] in various fields, they have been widely applied to time series anomaly detection methods by capturing the temporal correlations within the time series data using the hidden layers of Neural Networks (NNs). Although Recurrent Neural Networks (RNNs) [19] have achieved success in processing the sequential data, they exhibit the low computational efficiency when handling long sequences and are susceptible to some training issues, such as vanishing and exploding gradients. On the other hand, Long Short-Term Memory (LSTM) [20, 21, 22, 13] networks are able to mitigate the issue of vanishing gradients and capture long-term dependencies. Consequently, they have been employed in many unsupervised anomaly detection tasks. Nevertheless, these networks also have to suffer from the high computational cost.

Similarly, Transformer [23] can also be used to process the sequential data. Compared with LSTM, Transformer exhibits the higher parallelism, which results in the faster processing of long sequences, while it also excels in handling long-term dependencies. Therefore, Transformer has been applied to anomaly detection in the multivariate time series data, for example, the TranAD [24] and DTAAD [25] methods. However, these methods will struggle with the anomaly detection tasks performed in the edge computing scenarios which are prevalent in industrial production, due to the limited computational resource. In this situation, an efficient and effective time series anomaly detection network, in particular, tailored for edge devices on the manufacturing line, is demanded.

To this end, we propose a lightweight industrial time-series anomaly detection network, termed Efficient Periodicity-Aware Anomaly Detection Network (EPAAD-Net). The network adopts a streamlined encoder-decoder architecture built upon 1D convolutions, which considerably reduces computational overhead while preserving competitive detection performance. To strengthen the ability of the network to capture the inherent periodic patterns in industrial signals, we design a Sine-Cosine Transform Module (SCTM) that projects the extracted deep features onto periodic basis functions through learnable weighted sine and cosine mappings. This design enhances the sensitivity of the network to phase-sensitive anomalies associated with rotational frequencies and intermittent material feeding. In addition, a cross-attention module is inserted between the encoder and decoder to capture long-range dependencies across temporal positions. With these designs, EPAAD-Net achieves a favorable balance between detection accuracy and computational efficiency, making it particularly suitable for resource-constrained edge deployment in industrial environments.

The main contributions of this study can be summarized as follows.

(1) We construct a lightweight periodicity-aware anomaly detection network, i.e., EPAAD-Net, tailored for edge deployment on manufacturing lines. The network adopts a streamlined encoder-decoder architecture built on 1D convolutions, achieving low-latency inference on resource-constrained devices through a carefully balanced trade-off between model complexity and detection accuracy.

(2) We design a Sine-Cosine Transform Module (SCTM) that projects deep features onto periodic basis functions via learnable weighted sine and cosine mappings. This module serves as a periodic-prior embedding mechanism, enhancing the sensitivity of the network to the inherent periodic patterns in industrial signals, such as roller rotation frequencies and intermittent feeding rhythms, and consequently improving its ability to detect phase-sensitive anomalies.

(3) We conduct extensive experiments on six publicly available time-series anomaly detection data sets, systematically comparing against 14 baseline methods. The results demonstrate that EPAAD-Net achieves detection accuracy superior to, or at least comparable to, state-of-the-art baselines on most data sets, while offering particular advantages in inference latency over mainstream lightweight models.

It should be noted that the primary contribution of this work lies not in devising entirely novel mathematical components, but in the task-oriented integration and lightweight adaptation of existing well-established modules, such as 1D convolutions, autoencoders, and attention mechanisms, for industrial production scenarios, thereby enabling reliable and efficient anomaly detection under edge-deployment constraints.

The rest of this paper is organized as follows. We first review the relevant work in Section 2. The proposed EPAAD-Net is then introduced in Section 3 in detail. In Sections 4 and

5, we report the experimental setup and results, respectively. Finally, we draw the conclusion in Section 6.

2. Related Work

2.1. Traditional Methods

The detection of anomalies in time series data is key to industrial production processes. As the foundation of anomaly detection in time series data, traditional methods were typically designed based on the principles of statistics and classical machine learning algorithms. For example, K-means clustering [15, 26], Support Vector Machine (SVM) [12, 14], regression models [11], Principal Component Analysis (PCA) [7], and hidden Markov chains [8] have been commonly used for time series anomaly detection tasks. In addition, transform-based methods, such as Discrete Wavelet Transform (DWT) [27] and Fourier Transform (FT) [28], have been employed to process time series data to facilitate efficient anomaly detection.

Unsupervised time series anomaly detection methods have also been developed based on distance and density information [29, 30, 31]. [32] proposed an isolation forest anomaly detection algorithm that isolates anomaly points according to a set of rules and assesses them based on the number of partitions required. [33] improved the isolation forest algorithm by integrating K-means clustering and nearest neighbor approaches. [34] proposed a graph-based subsequence anomaly detection system (GraphAn) that transforms time series data into graphs and performs anomaly detection based on graph distance. Similarly, [35] introduced an anomaly detection method based on Graph Neural Networks (GNN) for IoT scenarios.

Although the aforementioned studies have made progress in detecting anomalies in time series, they often struggle with handling complex temporal patterns and large volumes of data. Due to the powerful representation learning ability of deep learning techniques, an increasing number of studies have adopted these advanced methods.

2.2. LSTM-Based Methods

Recently, the Long Short-Term Memory (LSTM) network [20] has been widely applied to time series anomaly detection. [21] proposed an unsupervised anomaly detection algorithm based on an LSTM Encoder-Decoder (LSTM-ED) architecture. [22] developed a scalable hybrid model for real-time anomaly detection in industrial control system network traffic by combining LSTM with Seasonal Autoregressive Integrated Moving Average (SARIMA) and dynamic thresholding. [13] utilized an LSTM network together with a one-class SVM to discriminate abnormal data in vibration signals. To capture temporal correlations in multivariate time series, [36] combined a Graph Convolutional Network (GCN) with LSTM and a Variational Autoencoder (VAE). [37] introduced a Deep Convolutional Autoencoding Memory (CAE-M) network that integrates a deep convolutional autoencoder with an autoregressive model and a bidirectional LSTM. Additionally, [38] constructed a multi-scale feature matrix and employed a convolutional encoder

combined with a ConvLSTM network to capture temporal patterns.

However, LSTM-based methods are often time-consuming when processing long time series data and typically involve a large number of parameters, resulting in high computational costs during training.

2.3. Transformer-Based Approaches

Compared with LSTM networks, Transformer networks generally have a simpler structure and fewer parameters. [39] proposed an anomaly detection transformer based on an anomaly-attention mechanism. Several studies have focused on multidimensional time series anomaly detection by integrating adversarial training with Transformer techniques [40, 41, 42]. [43] introduced a Transformer network incorporating dilated convolution to capture both local and long-term features at different dilation rates. [44] developed a Transformer-Based Self-feedback Network (TSFN) to enhance the learning of potential anomalies while capturing temporal dependencies. [45] designed a Bi-Transformer Anomaly Detection (BTAD) method that processes data from two different dimensions in parallel to improve detection efficiency.

Although the above-mentioned approaches achieve promising results, they often suffer from large model sizes, high computational costs, and long training times. Consequently, these methods face challenges when deployed at the edge of production lines.

2.4. Lightweight Network-Based Methods

To address the bottleneck of deploying anomaly detection networks on edge devices, lightweight network architectures have been explored. [46] reduced computational complexity and training time through a series of optimization mechanisms. [28] developed a Federated Learning-Based Autoencoder-Transformer-Fourier Anomaly Detection (FATRAF) method that replaces the traditional self-attention sublayer with a non-parameterized Fourier sublayer. [47] achieved a reduction in model size and training time by employing a simplified autoencoder structure. [48] designed a lighter model by utilizing only the encoder blocks of the transformer. [24] introduced TranAD, a deep Transformer-based anomaly detection method that amplifies and reconstructs errors through adversarial training. Although the Dual Temporal Convolutional Network (TCN) Attention Network for Anomaly Detection (DTAAD) [25] outperforms TranAD on some datasets, it requires more parameters.

2.5. Lightweight Anomaly Detection in Encrypted Network Environments

Beyond industrial manufacturing, recent advances in securing encrypted network communications have highlighted the growing demand for lightweight and privacy-preserving anomaly detection at the edge. Contemporary paradigms in encrypted Domain Name System (DNS) security, such as next-generation AI for advanced threat detection [49],

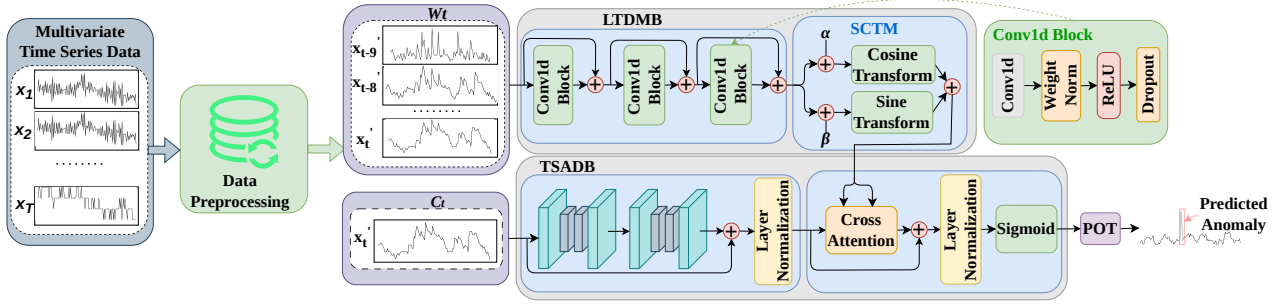


Figure 1: The architecture of the proposed Efficient Periodicity-Aware Anomaly Detection Network (EPAAD-Net).

quantum-enhanced autonomous frameworks [50], and meta-morphic intelligence architectures [51], have demonstrated the potential of AI-driven approaches to address the challenges posed by encrypted traffic, polymorphic tunneling, and adaptive adversarial behaviors. In addition, federated AI attestation frameworks [52] have been proposed to unify distributed learning with quantum-resilient cryptography and neuromorphic anomaly detection in a privacy-preserving manner, while evolutionary encryption schemes [53, 54] have introduced adaptive cryptographic parameter modulation to harden encrypted DNS communications against both classical and quantum-accelerated threats.

3. The Efficient Periodicity-Aware Anomaly Detection Network

During the industrial production process, a substantial volume of time series data will be continuously generated through edge devices. In this situation, fast and effective anomaly detection is demanded. Considering real-time constraints and limited resources available at the edge devices, however, it is still challenging to develop an efficient and effective time series anomaly detection method. To address this challenge, we propose a novel time series anomaly detection network, namely, Efficient Periodicity-Aware Anomaly Detection Network (EPAAD-Net). In particular, we design a Sine-Cosine Transform Module (SCTM) that introduces a periodic inductive bias by projecting the extracted features onto sine and cosine basis functions, enabling the network to better capture periodic patterns in the time-series data. For the sake of encoding the long-term dependencies between data points, a cross-attention module is also inserted into the EPAAD-Net. As a result, our EPAAD-Net derives a proper trade-off between the efficiency and effectiveness. To our knowledge, such a periodicity-aware anomaly detection network has not been explored in the previous studies. The architecture of the proposed EPAAD-Net is illustrated in Fig. 1.

3.1. Problem Definition

The time series data collected at the edge devices of the production line can be regarded as the multivariate time series data, which is a time-stamped sequence with the size

of T . The time series data can be expressed as:

$$X = \{x_1, \dots, x_T\}, \quad (1)$$

where X represents the entire time series data and x_t denotes the data point at a specific timestamp t . In other words, X comprises all the data points collected across the T timestamps. Given a timestamp t , $x_t \in \mathbb{R}^m$ holds true. When $m = 1$, in particular, it means that the time series data is univariate. We will define the problem of time series anomaly detection as follows.

Problem 1 Anomaly Detection. Regarding the time series data X obtained at the edge of the production line, which may consists of either the univariate data or the multivariate data, we use this data set as the training data to train the network. In terms of the testing data set $\hat{X}$, the anomaly detection task aims to reconstruct these data. The result is a set of reconstructed data $\hat{Y}$. The anomaly scores can be calculated based on the data set $\hat{Y}$. Whether or not the data point at timestamp t in the testing data set is anomalous can be determined by applying a threshold to these scores.

3.2. Data Preprocessing

Data preprocessing is of the paramount importance to time series anomaly detection tasks. A proper preprocessing operation enables the method to better capture actual data patterns and thereby lead to the better performance. Normally, the normalization operation can reduce data volatility and noise, which improves the stability and robustness of the anomaly detection method. We therefore use this technique to process the data before they are fed into the network. The normalization operation can be expressed as:

$$x'_t = \frac{x_t - \min(X)}{\max(X) - \min(X) + \delta}, \quad (2)$$

where $\min(X)$ and $\max(X)$ represent the vectors with the smallest and largest modes across the time series data, respectively, δ is a tiny constant vector which is used to avoid the zero-division incident. The time series data X is normalize to the range of $[0, 1)$ using the above formula. The normalized data is denoted as X' .

Considering that the occurrence of the anomaly in time series is usually not independent but is influenced by the context correlation, we are motivated to convert the normalized

time series data into a series of sliding windows [47, 24, 25], to capture the local dependencies in the time series data. Given the data point x_t' at timestamp t , a contextual window can be sampled as the L prior points (including this point), which can be expressed as

$$W_t = \{x_{t-L+1}', \dots, x_t'\}. \quad (3)$$

When $t < L$, we use the data points at timestamp t to generate a uniform window instead. As a result, the normalized time series $X' = \{x_1', \dots, x_T'\}$ is transformed into a sliding window sequence $W = \{W_1, \dots, W_T\}$. In the rest of this paper, we denote x_t' as C_t . Both C_t and W_t ($\hat{C}_t$ and $\hat{W}_t$ in the testing process) are fed into the network rather than C_t ($\hat{C}_t$ in the testing process) only during the training and testing processes. In this case, the model trained is able to perform the anomaly detection task on top of both the short-term and long-term dependencies, which not only reduces the fluctuations in the anomaly scores that the model predicts but also improves the performance of the model.

Specifically, we followed the preprocessing protocol from the original publication [36], which includes normalizing each dimension to zero mean and unit variance, and partitioning the data into non-overlapping training and test sets according to the original entity splits, for the SMD [55] data set. For the NAB [56] data set, we removed the erroneously labeled anomaly sequences as identified in the previous work [24], and applied min-max normalization to the univariate time series. The other data sets that we used in this study were preprocessed according to their original publications, as detailed in Section 4.1.

3.3. EPAAD-Net

Fig. 1 shows the architecture of the proposed EPAAD-Net. Motivated by the previous study [24], this network is built on top of the dual-branch architecture, which consists of a Long-Term Dependency Modeling Branch (LTDMB) and a Target Sequence Anomaly Detection Branch (TSADB). The data point at the current timestamp t and the data points within the corresponding sliding window are simultaneously fed into the EPAAD-Net for inference. The two sets of data points are denoted as C_t and W_t , respectively.

Compared with the TranAD method introduced in [24], EPAAD-Net is designed on top of a lightweight design. This design discards the self-attention module, which uses a large number of parameters, but uses one-dimensional convolution, autoencoder and the SCTM that we deliberately propose instead. In addition, we do not use adversarial techniques, which will decrease the inference speed of the model.

The dual-branch architecture decouples two complementary tasks, i.e., global context modeling and local anomaly detection, using the LTDMB and TSADB, respectively. Self-attention is omitted due to its $O(n^2)$ complexity, which imposes excessive latency on edge device. Instead, 1D convolutions with the $O(n)$ complexity are used in the LTDMB to capture local temporal correlations efficiently. Cross-attention is retained in the TSADB to query relevant context from the LTDMB with the $O(n)$ complexity when the

query length is fixed, enabling selective attention without the quadratic cost. This two-branch design is motivated by the distinct objectives of the two tasks, allowing specialization without interference, while cross-attention coordinates information flow.

3.3.1. LTDMB

Although existing deep networks may achieve proper performance, they normally use a large number of parameters. To address this issue, the LTDMB is designed with the objective of obtaining a lightweight implementation. To be specific, the LTDMB comprises three one-dimensional convolutional (Conv1d) blocks and a Sine-Cosine Transform Module (SCTM). The Conv1d blocks are used to capture the correlation between time series data, while the SCTM is designed in order to enhance the periodicity of these data. Since we only utilize three Conv1d blocks and an SCTM, the number of the parameters contained in the LTDMB is relatively small.

Conv1d Blocks: To capture the important information contained in the input sequence data, while keeping the network as lightweight as possible and maintaining a stable and controllable network structure, we use Conv1d blocks with the same dimension for both the input and output data. In total, three continuous Conv1d blocks are stacked. Each block contains a one-dimensional convolutional layer, a weight normalization layer, a ReLU activation function and a dropout layer. The utilization of weight normalization reduces the possibility of gradient vanishing or exploding because it helps the training of the network converge faster. As a nonlinear activation function, the ReLU function enables the network to learn the more complex patterns. Since the dropout layer randomly sets a portion of neurons to zero, it reduces the interdependence between neurons. As a result, the model trained is prevented from over-fitting the training data and the generalization ability of the model is enhanced.

Regarding each Conv1d block, the residual connection is used. That is to say that the input of each Conv1d block is added to the output of this block. The sum is then used as the input of the next block. Given that the input is denoted as W_i , the convolution operations can be expressed as:

$$W_i' = W_i + \text{Dropout}(\text{ReLU}(\text{WeightNorm}(\text{Conv1d}(W_i)))), \quad (4)$$

$$W_i'' = W_i' + \text{Dropout}(\text{ReLU}(\text{WeightNorm}(\text{Conv1d}(W_i')))), \quad (5)$$

$$W_i^1 = W_i'' + \text{Dropout}(\text{ReLU}(\text{WeightNorm}(\text{Conv1d}(W_i'')))). \quad (6)$$

SCTM: The Fourier series representation is based on the principle that a periodic signal can be decomposed into

a superposition of sine and cosine components of different frequencies [57]. Given a periodic signal $f(t)$ with period T , its Fourier series is expressed as:

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\omega t) + b_n \sin(n\omega t)), \quad (7)$$

where a_0 is the constant coefficient, ω is the angular frequency, n denotes the harmonic number, and a_n and b_n are the coefficients of the cosine and sine components, respectively. Inspired by the principle, we design a lightweight feature projection module, namely, Sine-Cosine Transform Module (SCTM), which maps the deep features extracted by the network onto a space spanned by sine and cosine basis functions, thereby introducing a periodic inductive bias into the network.

It should be noted that SCTM draws inspiration from the combination of sine and cosine basis functions rather than performing explicit Fourier decomposition. Let $\mathbf{h} \in \mathbb{R}^d$ denote the input to the module (i.e., the output of the last Conv1d block). The computation of SCTM is formulated as:

$$\mathbf{W}_i^2 = \epsilon \cdot \cos(\alpha \odot \mathbf{W}_i^1) + (1 - \epsilon) \cdot \sin(\beta \odot \mathbf{W}_i^1), \quad (8)$$

where $\odot$ denotes element-wise multiplication, $\alpha, \beta \in \mathbb{R}^d$ are learnable frequency-scaling vectors, and $\epsilon \in (0, 1)$ is a learnable weighting coefficient. The learnable vectors α and β are d -dimensional vectors initialized with Xavier uniform.

From a computational perspective, SCTM applies sine and cosine basis transformations to the features and then combines them with a learnable weight. However, its design intention is not to propose a mathematically novel transform, but rather to encode periodic priors into the feature representation with minimal parameter overhead. The two vectors α and β are learned individually, allowing the module to independently adjust the frequency responses of the sine and cosine components, and thus adapt to potential phase shifts or asymmetric periodic patterns in the input signals. The paired sine and cosine projections preserve phase information, enabling the network to distinguish anomalies with identical amplitudes but different phase characteristics, a type of anomaly that carries clear physical meaning in rotary machinery-dominated industrial processes.

The rationale behind SCTM is to enhance the detection of phase anomalies in periodic industrial signals. Anomalies can be broadly divided into two types: amplitude-type, where the signal deviates from its normal range, common in both periodic and non-periodic data and detectable with original features, and phase-type, where the signal remains within normal amplitude but exhibits temporal deviations. The latter is particularly prevalent in rotary machinery and is difficult to capture without explicit periodic priors. SCTM addresses this issue by projecting features onto the sine and cosine basis functions, improving sensitivity to phase-related anomalies, while the dual-branch architecture preserves the original feature representation to ensure that amplitude-type detection remains unaffected.

Recent studies, such as Fourier feature mapping [58] and SIREN [59], also employ sine and/or cosine transformations but differ from SCTM in objective and implementation. Fourier feature mapping applies fixed sinusoidal embeddings to input coordinates for learning high-frequency functions, while SIREN uses sine as a periodic activation function with fixed frequency ω_0 for continuous signal representation. In contrast, SCTM is a feature projection module inserted after a convolution, not an input embedding or activation. Unlike these fixed-frequency approaches, SCTM introduces learnable frequency-scaling vectors α and β , enabling adaptive frequency response. Besides, it is specifically designed for industrial time-series anomaly detection, where the goal is to enhance the separability between normal periodic patterns and phase-sensitive anomalies rather than signal reconstruction.

To intuitively compare the effects of the Fourier module [28] and the proposed SCTM on processing the time series data, we plot the time series features processed by both modules in Fig. 2. It can be observed that the features processed by the SCTM exhibit more regular periodic patterns, and the separation between abnormal and normal subsequences appears to be more distinct. This observation can be attributed to the learnable frequency-scaling mechanism of α and β in the SCTM, which adaptively adjusts the frequency responses to the data. These results qualitatively suggest that the proposed SCTM offers promising discriminative advantages over the Fourier attention module in the context of time-series anomaly detection. This observation is further supported by the ablation experiment reported in Section 5.4.1, where removing the SCTM leads to consistent performance degradation across all six data sets, confirming its positive contribution to anomaly discrimination.

On time series data with weak or without global periodicity, the gain from the SCTM is modest but not detrimental, because most anomalies are amplitude-type and the original features already provide sufficient information. The periodic-view features serve as a supplementary cue rather than a replacement. As shown in our ablation study (Section 5.4.1), removing the SCTM causes performance degradation across all evaluated data sets, including those with weak periodic patterns, with more pronounced effects on strongly periodic ones. This finding confirms that the SCTM provides complementary information when periodic structures exist but is not harmful even when they are absent.

3.3.2. TSADB

Within the TSADB, a denoising module and an anomaly prediction module are included. The denoising module aims to improve the quality of the data and enhance detection accuracy by alleviating the interference of the noise. The output of this module interacts with the output of the LT-DMB through the cross-attention mechanism, enabling the network to dynamically balance local features and global dependencies. As a result, high-quality feature representations can be obtained for the subsequent anomaly scoring module.

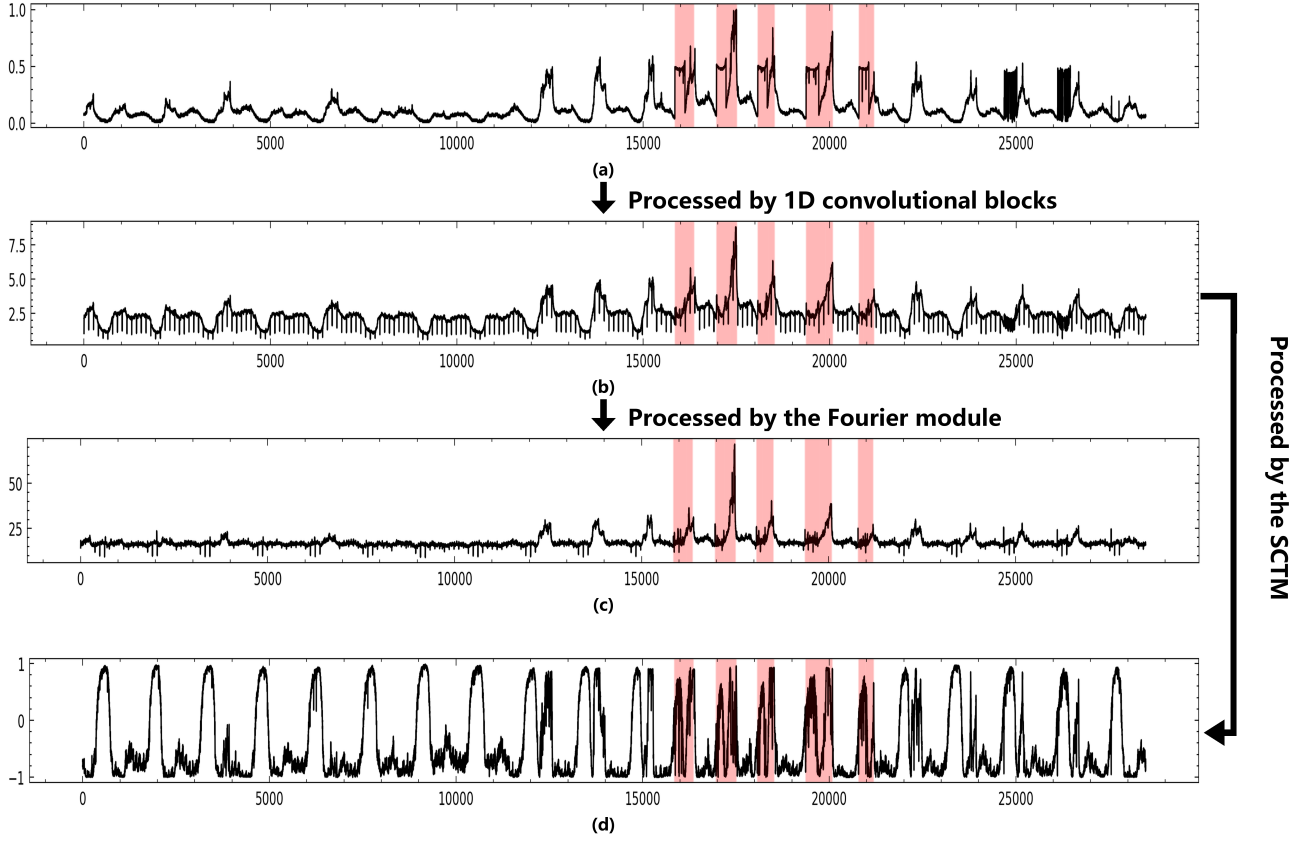


Figure 2: Comparison of the time-series features processed by the Fourier attention module [28] and the proposed SCTM. (a) Original window data. (b) Features extracted by the Conv1d blocks in the LTDMB (see Fig. 1). (c) Features after processing by the Fourier attention module. (d) Features after processing by the proposed SCTM. The pink background indicates that the sub-sequence at the associated timestamp is an anomaly. It can be observed that the SCTM-processed features exhibit more distinct separation between normal and abnormal sub-sequences compared to those processed by the Fourier attention module.

The denoising module contains a stacked encoder-decoder with the residual connection and a layer normalization operation. Let the time series data sent to the TSADB be denoted as C_i , the processing of the denoising module can be expressed as:

$$C_i^1 = \text{LayerNorm}(C_i + \text{ED}(\text{ED}(C_i))), \quad (9)$$

where $\text{ED}(\text{ED}(\cdot))$ represents the stacked encoder-decoder and $\text{LayerNorm}(\cdot)$ denotes the layer normalization operation. The encoder-decoder consists of two autoencoders (including linear layers with ReLU and residuals), each compressing d hidden dimension to $d/3$ hidden dimension and restoring them to d hidden dimension. The cross-attention uses $h = d$ heads, followed by a Feed-Forward Network (FFN) with a hidden size of 16 and LeakyReLU. Dropout is set to 0.1 with residual connections.

The output of the denoising module is then combined with that of the LTDMB using the cross-attention mechanism. Furthermore, the residual connection and a layer normalization operation are applied to the results of the cross-attention block. The above computation can be formulated

as:

$$C_i^2 = \text{LayerNorm}(C_i^1 + \text{CrossAtt}(C_i^1, W_i^2, W_i^2)), \quad (10)$$

where $\text{CrossAtt}(C_i^1, W_i^2, W_i^2)$ stands for the cross-attention operation and $\text{LayerNorm}(\cdot)$ denotes the layer normalization operation.

The cross-attention module follows the standard multi-head attention mechanism. We use the encoded features C_i^1 generated by the TSADB as the Query, and the output of the LTDMB, W_i^2 , as the Key and Value. The number of attention heads is configured in order that each head has a dimension of $d_k = d/h$, where d is the dimension of input features and h is the number of heads. The output dimension remains d after concatenation and the output projection, following the standard formulation. Structural modifications are not made to the standard cross-attention. The only difference is the omission of the scaling factor $1/\sqrt{d_k}$, which slightly reduces computational overhead with a negligible impact on performance.

Finally, the output of the layer normalization operation is fed into the Sigmoid activation function, which can be

Algorithm 1 The training operation of the proposed EPAAD-Net

Input:

Training Set W, C
Number of Iterations N

```

1: Initialize weights
2:  $n \leftarrow 0$ 
3: do
4:   for( $i = 1$  to  $T$ )
5:      $W'_i = W_i + \text{Dropout}(\text{ReLU}(\text{WeightNorm}(\text{Conv1d}(W_i))))$ 
6:      $W''_i = W'_i + \text{Dropout}(\text{ReLU}(\text{WeightNorm}(\text{Conv1d}(W'_i))))$ 
7:      $W^1_i = W''_i + \text{Dropout}(\text{ReLU}(\text{WeightNorm}(\text{Conv1d}(W''_i))))$ 
8:      $W^2_i = \text{SCTM}(W^1_i)$ 
9:      $C^1_i = \text{LayerNorm}(C_i + \text{ED}(\text{ED}(C_i)))$ 
10:     $C^2_i = \text{LayerNorm}(C^1_i + \text{CrossAtt}(C^1_i, W^2_i, W^2_i))$ 

11:     $O_i = \text{Sigmoid}(C^2_i)$ 
12:     $L = \|O_i - C_i\|_2$ 
13:    Use  $L$  to update the weights
14:     $n \leftarrow n + 1$ 
15: while  $n < N$ 

```

expressed as:

$$O_i = \text{Sigmoid}(C^2_i). \quad (11)$$

The result is the reconstructed version of the input time series data C_i , which will be used to compute the reconstruction error for anomaly detection.

3.4. Offline Model Training

Considering the Mean Squared Error (MSE) metric has been widely utilized in various time series anomaly detection tasks, we use it as the loss function of the proposed EPAAD-Net. The calculation of the MSE loss is performed based on the difference between the predictions of the model and the actual values. This loss function can be expressed as:

$$L = \frac{1}{N} \sum_{i=1}^N (O_i - C_i)^2, \quad (12)$$

where N is the number of samples, C_i is the actual value and O_i is the value predicted using the model. The goal of the training operation is to minimize the loss L . The training operation of the EPAAD-Net, is shown in Algorithm 1, in which $\text{Conv1d}(\cdot)$ represents the one-dimensional convolution block and $\text{CrossAtt}(\cdot)$ stands for the cross-attention module.

3.5. Online Detection

Once the training operation has been complete, a lightweight model will be obtained. This model can be used for time series anomaly detection. It should be noted that we do

not directly use the time series data for anomaly detection. Instead, the sliding window data is used in the anomaly detection task. The anomaly score is computed as the reconstruction error. We apply a thresholding process to the anomaly score. To be specific, the corresponding data point is treated as an anomaly if a score is greater than the threshold; otherwise, the data point is regarded as the normal data.

We use the Peak Over Threshold (POT) [60] method to automatically select a threshold. The rationale behind this choice is that anomalies, by definition, correspond to extreme values in the reconstruction error distribution, while POT, as an extreme value theory approach, is therefore naturally suited for modeling the upper tail of the error scores to identify anomalous points. We hypothesize that the associated data point should be marked as anomalous if one or more of the m dimensions in the time series data exhibit an anomaly at the current timestamp. This processing can be formulated as:

$$y_i = 1(s_i \geq \text{POT}(s_i)),$$

$$y = \begin{cases} 1, & \exists y_i = 1, \\ 0, & \text{otherwise,} \end{cases} \quad (13)$$

where i represents the current dimension, s_i denotes the anomaly score of the current dimension and y stands for the anomaly label.

The testing process of the EPAAD-Net can be described in Algorithm 2. In Fig. 3, we plot the input sub-sequence data, the sub-sequence data predicted using our model and the associated anomaly scores, along with the anomaly detected using our method. As can be seen, the proposed EPAAD-Net is able to reconstruct the original data with some errors and the anomaly can be detected by applying the POT method to the anomaly scores.

4. Experimental Setup

In this section, we will introduce the experimental setup, including the data sets, baselines and performance metrics utilized in our experiments and the implementation details of the proposed approach.

4.1. Data Sets

We used six publicly available data sets, which had been widely used as the benchmark in different time series anomaly detection tasks. Table 1 lists the details of these data sets, including the name of the data set, the size of the training set, the size of the testing set, the number of entities, the dimensions (Dim.) of the data and the percentage (PCT) of the anomalies contained in each testing set. We will briefly describe each of the six data sets below.

1. *Server Machine Data Set (SMD)*: The SMD [55] data set is normally used for server fault detection, which records the operations and failure events of servers in the real world throughout five weeks. This data set comprises the multivariate time series data with 38

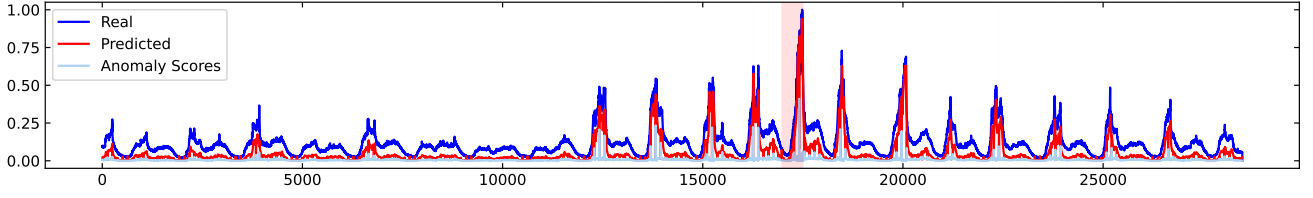


Figure 3: Examples of the real sub-sequence data, the sub-sequence data predicted by the EPAAD-Net model and the associated anomaly scores. Here, the anomaly detected is indicated using the pink background.

Algorithm 2 The testing operation of the EPAAD-Net

Input:

Testing Set $\hat{W}, \hat{C}$

- 1: **for** ($i = 1$ to T)
- 2: $\hat{W}'_i = \hat{W}_i + \text{Dropout}(\text{ReLU}(\text{WeightNorm}(\text{Conv1d}(\hat{W}_i))))$
- 3: $\hat{W}''_i = \hat{W}'_i + \text{Dropout}(\text{ReLU}(\text{WeightNorm}(\text{Conv1d}(\hat{W}'_i))))$
- 4: $\hat{W}^1_i = \hat{W}''_i + \text{Dropout}(\text{ReLU}(\text{WeightNorm}(\text{Conv1d}(\hat{W}''_i))))$
- 5: $\hat{W}^2_i = \text{SCTM}(\hat{W}^1_i)$
- 6: $\hat{C}^1_i = \text{LayerNorm}(\hat{C}_i + \text{ED}(\text{ED}(\hat{C}_i)))$
- 7: $\hat{C}^2_i = \text{LayerNorm}(\hat{C}^1_i + \text{CrossAtt}(\hat{C}^1_i, \hat{W}^2_i, \hat{W}^2_i))$
- 8: $\hat{O}_i = \text{Sigmoid}(\hat{C}^2_i)$
- 9: $s_i = \|\hat{O}_i - \hat{C}_i\|_2$
 $y_i = 1(s_i \geq \text{POT}(s_i))$
- 10: $y = \begin{cases} 1, & \exists y_i = 1 \\ 0, & \text{otherwise} \end{cases}$

dimensions. We used the non-trivial sequences in the SMD data set.

2. *Numenta Anomaly Benchmark (NAB)*: The NAB [56] data set covers the univariate time series data captured in different fields, including energy, network traffic, industrial production. This data set is commonly used for anomaly detection in time series. We removed the erroneous labeled anomaly sequences contained in the NAB data set.
3. *MIT-BIH Supraventricular Arrhythmia Database (MBA)*: The MBA [61] data set contains the electrocardiogram data collected from different patients, which is the multivariate time series data with the dimensionality of 2. Two different types of anomalies are included in this data set.
4. *Soil Moisture Active Passive (SMAP) Data Set*: The SMAP [62] data set comprises the remote sensing data collected by the Mars probe of NASA. These multivariate time series data has the dimensionality of 25.
5. *Secure Water Treatment (SWaT_MV) Data Set*: The SWaT_MV [63] data set was derived by continuously

Table 1

The details of the data sets used in this study.

Data Set	Training	Testing	Entities	Dim.	PCT
SMD [55]	708,420	708,420	4	38	4.16
NAB [56]	4,033	4,033	6	1	0.92
MBA [61]	100,000	100,000	8	2	0.14
SMAP [62]	135,183	427,617	55	25	13.13
SWaT_MV [63]	496,800	449,919	1	51	11.98
SWaT_UV [24]	3000	5000	1	1	12.88

running a water treatment system for seven days of normal operations and four days of abnormal operations. The dimensionality of these multivariate time series data is 51, which can be used for industrial control system security research. Due to the complex industrial environment and multivariate temporal dependencies, however, the SWaT_MV data set is challenging.

6. *Secure Water Treatment (SWaT_UV) Data Set*: The SWaT_UV [24] data set was obtained by processing the SWaT_MV [63] data set. This data set contains the univariate time series data.

4.2. Baselines

In total, we used three traditional and 11 deep learning-based baseline methods. We will briefly introduce these methods as follows.

1. *One-Class Support Vector Machine (OCSVM)* [64]: This approach relies on the traditional SVM classifier, which is trained to discern the boundary between the normal and abnormal data points.
2. *Local Outlier Factor (LOF)* [65]: The LOF method is an outlier detection method based on the degree of isolation of an object in its local neighborhood. This method assigns a degree of being an outlier to each object.
3. *Isolation Forest* [66]: This approach is an outlier detection method on top of the idea of isolation, which fulfils anomaly detection by explicitly isolating anomalies.
4. *Omnianomaly* [55]¹: To capture the normal patterns of the multivariate time series data, this method learns the robust representations of these data by means of

¹<https://github.com/NetManAIops/OmniAnomaly>

the stochastic variable connection and planar normalizing flow techniques.

5. *Multi-scale Convolutional Recurrent Encoder-Decoder (MSCRED)* [38]²: The MSCRED encodes the correlations between the time series data using a convolutional encoder and captures temporal patterns through the attention-based Convolutional Long Short-Term Memory (ConvLSTM) network.
6. *Multivariate Anomaly Detection with GAN (MAD-GAN)* [67]³: This method uses the Generative Adversarial Network (GAN) built on the LSTM-RNN model to capture the temporal correlation of the time series distribution. Both the generator and discriminator of the GAN can be used to detect anomalies based on new anomaly scores.
7. *Unsupervised Anomaly Detection for Multivariate Time Series (USAD)* [47]: The USAD approach utilizes an autoencoder with two decoders and an adversarial training framework. This approach is one of the first studies which reduce the overhead using a simple autoencoder.
8. *Multivariate Time-Series Anomaly Detection via Graph Attention Network (MTAD-GAT)* [68]: As a multivariate time series anomaly detection method, the MTAD-GAT method was built on top of the graph attention network. This method introduced the prediction and reconstruction models for the joint optimization.
9. *Deep Convolutional Autoencoding Memory Network (CAE-M)* [37]: The CAE-M method captures the time correlation from the time series data using a memory network which comprises both the linear (autoregressive model) and non-linear (bidirectional LSTM with attention) prediction networks.
10. *Graph Deviation Network (GDN)* [69]: This method combines the structural learning method with the graph neural network. Attention weights were also used to provide the interpretability for the anomalies detected.
11. *Transformer-Based Anomaly Detection Model (TranAD)* [24]⁴: To capture the temporal trends contained in the series data, the TranAD approach was built on top of an attention-based sequence encoder. A framework designed based on the focus score was also developed for self-adjustment and adversarial training.
12. *Dual Temporal Convolutional Network Attention Network for Anomaly Detection (DTAAD)* [25]⁵: The DTAAD method is a lightweight anomaly detection method, which was designed on the basis of the transformer and dual Temporal Convolutional Network (TCN). For the purpose of improving the prediction accuracy and expanding the correlation differences, both the scaling method and the feedback mechanism were introduced into this method.
13. *TimesNet* [70]: A general time-series analysis backbone that transforms 1D time series into 2D tensors based on multiple periods discovered by the Fourier transform, and employs 2D convolutions to capture complex temporal variations. It was tested in five mainstream tasks, such as anomaly detection.
14. *DCdetector* [71]: A multi-scale dual attention contrastive representation learning model for time series anomaly detection. It employs a novel asymmetric dual attention design with pure contrastive loss to learn permutation-invariant representations with superior discrimination abilities.

For the baselines whose source code had been made publicly available, including Omnianomaly [55], MSCRED [38], MAD-GAN [67], TranAD [24], DTAAD [25], TimesNet [70] and DCdetector [71], we used the original implementation. For the baselines whose source code was not made publicly available in the original publication, including USAD [47], MTAD-GAT [68], CAE-M [37] and GDN [69], we used the re-implementation in [24]. No matter whether the original implementations or the re-implementations, we always used their original experimental settings in order to avoid the bias introduced by tuning the parameters and obtain as fair as possible comparative results.

To ensure a fair comparison in detection accuracy, we adopted the data splits and preprocessing protocols in the original publications for each baseline. Efficiency metrics were measured in identical hardware configurations, and all models were evaluated in the same framework to minimize implementation biases. It should be noted that threshold selection strategies may vary across baseline implementations, which could slightly affect precision-recall trade-offs. Since inference latency metrics are threshold-independent, efficiency comparison remains unaffected. In addition, runtime measures were computed using the original implementations of each baseline at the identical batch size and input length settings to ensure fairness.

In our experiments, USAD, MTAD-GAT, CAE-M and GDN were evaluated using third-party re-implementations, while EPAAD-Net used dataset-specific learning rates. To mitigate possible bias, we adopted the re-implementations from the widely-used codebase of TranAD [24], and set the learning rate of EPAAD-Net following the same dataset-specific strategy as that TranAD used, ensuring comparable optimization conditions with the strongest baseline.

It should be noted that FATRAF [28] was not included in the quantitative evaluation even if it was qualitatively compared in Fig. 2. The reason is that FATRAF is a federated learning-based framework designed for distributed deployment across multiple edge nodes, which requires a fundamentally different experimental setup (e.g., multiple clients, communication rounds and non-IID data partitions) that is beyond the scope of this study.

4.3. Performance Metrics

To evaluate the performance of an anomaly detection method, we used Precision, Recall, F1 Score and the Area

²<https://github.com/7fantasysz/MSCRED>

³<https://github.com/LiDan456/MAD-GANs>

⁴<https://github.com/imperial-qore/TranAD>

⁵<https://github.com/Yu-Lingrui/DTAAD>

under the Curve (AUC) as performance metrics by following the previous work. The Precision, Recall and F1 Score metrics can be computed as follows:

$$Precision = \frac{TP}{TP + FP}, \quad (14)$$

$$Recall = \frac{TP}{TP + FN}, \quad (15)$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}, \quad (16)$$

where TP is the number of true positives, FP is the number of false positives and FN is the number of false negatives.

4.4. Implementation Details

This study aims to develop a method which is able to rapidly and effectively detect anomalies in the resource constrained environment. Therefore, we only used the CPU for experimentation. In other words, both the training and testing operations of the proposed EPAAD-Net were performed on a PC, which had the configuration of the 11th Gen Intel (R) Core (TM) i5-11400 @ 2.60GHz 2.59 GHz, 8GB RAM and Windows 11 operating system.

We used the AdamW [72] optimizer to train the EPAAD-Net, with the initial learning rate of 0.01, the weight decay coefficient of $1e-5$, the learning rate scheduler step of 5 and the learning rate decay factor of 0.9. Regarding the hyperparameters, the window size, the number of one-dimensional convolutional layers, the number of autoencoders and ϵ (see Equation (8)) were set to 10, 3, 2 and 0.3, respectively. Different learning rates were set with regard to the six data sets, following the setup used in [24]. For the SMD [55], NAB [56], MBA [61], SMAP [62], SWaT_MV [63] and SWaT_UV [24], the learning rate was set to $1e-4$, 0.009, 0.001, 0.001, 0.008 and 0.008, respectively. In terms of the autoencoder, the number of hidden units was set to one third of the dimensionality of the data set.

The hyperparameters of our method, including the sliding window size L , the weighting coefficient ϵ , and the number of Conv1d blocks, were fixed throughout the experiments. Specifically, the window size was set to $L = 10$ following the common practice in previous studies [73] to balance the need for sufficient historical context and memory limitation. The coefficient ϵ in the SCTM was set to 0.3 as an empirical starting point that evenly balances the sine and cosine projections. The number of Conv1d blocks was set to 3 as a lightweight design choice, which is a trade-off between representational capacity and computational efficiency. In particular, one or two layers may not sufficiently capture local temporal patterns, while four or more layers may introduce excessive parameters with marginal performance gains. In addition, the learnable parameters α and β were optimized during the training process, and thus their values adapted to the data distribution rather than being pre-specified.

5. Experimental Results

Both the method introduced in Section III and the experimental setup described in Section IV were utilized in the main experiments. In addition, we performed a series of ablation experiments. In this section, we report the results derived in these experiments.

5.1. Main Experiments

5.1.1. Quantitative Evaluation

The six data sets shown in Table 1 were utilized in our main experiments. The proposed EPAAD-Net was compared with 14 time series anomaly detection baselines. Table 2 shows the F1 score and AUC values obtained using EPAAD-Net and the baselines on the six data sets. As can be seen, EPAAD-Net achieves competitive detection performance across the six data sets and two performance metrics. Compared to the two strongest baselines, i.e., TranAD and DTAAD, EPAAD-Net produces higher F1 scores on SMD, SMAP, and SWaT_MV, comparable results on NAB and SWaT_UV, and a slightly lower score on MBA. It is noteworthy that EPAAD-Net outperforms TranAD by 12.3% in F1 score on the most challenging data set with 51 dimensions, namely, SWaT_MV. Among the baselines, both TranAD [24] and DTAAD [25] perform better than the other baselines in most cases.

The average F1 score value produced by our method across the six data sets was 0.9272. Compared with the top two baselines, i.e., TranAD [24] and DTAAD [25], our method achieved the improvements of 2.8% and 3.8%, respectively, in this metric. The advantage of the proposed EPAAD-Net over its counterparts should be due to the fact that the periodicity of the time series data has been enhanced by the proposed SCTM, which is beneficial to encoding the correlation between these data using our method. It should be noted that the AUC value of 0.9510 produced by EPAAD-Net is slightly lower than that (0.9747) CAE-M achieves even though it yields the highest F1 score on SWaT_MV. This observation indicates that CAE-M exhibits a better ranking of anomaly scores on this data set, even if EPAAD-Net produces a higher F1 value.

Furthermore, EPAAD-Net, TranAD, and DTAAD yield identical F1 and AUC values (to four decimal places) on the NAB data set. This finding may be due to two reasons. First, these methods were adopted from the same evaluation protocol, including point-adjustment and POT thresholding. Second, a ceiling effect took place on this relatively simple univariate data set, where those methods achieve near-saturated performance.

For the purpose of performing a more discriminative comparison, we report point-wise F1 score values without a post-hoc point-adjustment (PA) in Table 3. Under this evaluation protocol, EPAAD-Net achieves the highest F1 scores on SMD, NAB, SWaT_MV and SWaT_UV, and remains competitive with the strongest baselines on MBA and SMAP. These results confirm that the superiority of EPAAD-Net is consistent across both PA and point-wise protocols.

Table 2

The performance of the EPAAD-Net and 14 baselines on six different data sets. Given a data set and performance metric, the best result is indicated in the **red, bold** font while the second best result is highlighted in the *blue, italic* font.

Method	SMD		NAB		MBA		SMAP		SWaT_MV		SWaT_UV	
	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC	F1	AUC
OCSVM (*)	0.5627	–	–	–	–	–	0.5634	–	0.4955	–	–	–
LOF (*)	0.4689	–	–	–	–	–	0.5552	–	0.6860	–	–	–
Isolation Forest (*)	0.5364	–	–	–	–	–	0.5553	–	0.4702	–	–	–
OmniAnomaly	0.9401	<i>0.9946</i>	0.7442	0.8330	0.9235	0.9576	0.8894	0.9880	0.8122	0.8667	0.8131	<i>0.8467</i>
MSCRED	0.8414	0.9921	0.7442	0.8330	0.9623	0.9799	0.8664	0.9821	0.8436	0.9422	0.8072	0.8433
MAD-GAN	0.9150	0.9933	0.7752	0.8478	0.9689	0.9836	0.8985	0.9891	0.8461	0.8883	0.8065	0.8456
USAD	0.9495	0.9933	0.7442	0.8330	0.9448	0.9701	0.8974	0.9890	0.8790	0.8931	<i>0.8143</i>	0.8438
MTAD-GAT	0.8663	0.9829	0.7804	0.8221	0.9480	0.9719	0.8880	0.9844	0.8535	0.9359	0.8109	0.8464
CAE-M	0.9367	0.9783	0.7968	0.8019	0.9374	0.9658	0.8827	<i>0.9901</i>	<i>0.9197</i>	0.9747	0.8101	0.8464
GDN	0.8342	0.9924	<i>0.7998</i>	<i>0.8542</i>	0.9332	0.9528	0.8518	0.9864	0.7995	0.8525	0.8101	0.8462
TranAD	<i>0.9502</i>	0.9934	0.9412	0.9996	<i>0.9780</i>	<i>0.9885</i>	0.8915	0.9883	0.8341	0.8775	<i>0.8143</i>	0.8438
DTAAD	0.9144	0.9892	0.9412	0.9996	0.9793	0.9892	<i>0.8996</i>	0.9892	0.8156	0.8511	0.8109	0.8463
TimesNet	0.8331	0.3242	0.6957	0.6966	0.9024	0.5963	0.8583	0.5245	0.8007	0.7830	0.8000	0.8136
DCdetector	0.9095	0.3279	0.4571	0.5670	0.5785	0.5600	0.8558	0.5696	0.7308	0.7618	0.7323	0.7818
EPAAD-Net (Ours)	0.9944	0.9982	0.9412	0.9996	0.9548	0.9758	0.9200	0.9916	0.9364	<i>0.9510</i>	0.8162	0.8502

Table 3

Point-wise F1 scores (without point adjustment) for all methods across six datasets, the best result is indicated in the **red, bold** font while the second best result is highlighted in the *blue, italic* font.

Method	SMD	NAB	MBA	SMAP	SWaT_MV	SWaT_UV
OmniAnomaly	0.3232	0.4000	0.2538	0.0514	0.2183	<i>0.7518</i>
MSCRED	0.3310	<i>0.4167</i>	0.1818	0.0603	0.2183	0.7475
MAD-GAN	0.3221	0.3784	0.2603	0.0514	0.2183	0.7484
USAD	<i>0.3318</i>	0.3590	0.2505	0.0515	0.2183	0.7399
MTAD-GAT	0.2687	0.4000	0.1809	0.1509	0.2183	0.7429
CAE-M	0.3246	0.3590	0.2450	0.0476	0.2183	0.7435
GDN	0.2417	0.3590	0.2521	0.0536	0.2183	0.7429
TranAD	0.3254	0.3889	<i>0.2573</i>	0.0617	<i>0.2183</i>	0.7070
DTAAD	0.2763	0.3590	0.1672	0.0318	0.2126	0.7239
TimesNet	0.3123	0.3902	0.2204	0.0639	0.2183	0.7240
DCdetector	0.1808	0.2857	0.2050	0.0656	0.2183	0.7435
EPAAD-Net (Ours)	0.3994	0.4667	0.2441	<i>0.0985</i>	0.2197	0.7552

5.1.2. Qualitative Evaluation

To provide a more intuitive comparison, we visualized the time series data in the SWaT_MV [63] data set processed by the TranAD [24] and our method. Fig. 4 shows the input data and its true labels, as well as the prediction results of two methods for abnormal positions in the data set. It can be observed that our method produced the better results than those obtained using TranAD on this data set. However, both methods had undetected anomalies, demonstrating the inherent challenges of the SWaT_MV data set.

5.2. Performance Analysis

Since the goal of this study is to develop a computationally efficient time series anomaly detection approach suitable for edge deployment, we evaluated both model complexity (number of parameters) and inference latency.

5.2.1. Model Complexity

As shown in Table 4, EPAAD-Net consistently uses fewer parameters than Transformer-based methods, such as TranAD and DTAAD, across all data sets. Compared to simpler autoencoder-based baselines, including OmniAnomaly, MAD-GAN, USAD, CAE-M and GDN, EPAAD-Net uses more parameters on the SMD and SMAP data sets, but utilizes fewer or comparable parameters on the remaining data sets. This reflects a design trade-off in which the additional parameters in the dual-branch architecture and cross-attention module contribute to improving accuracy on challenging data sets, while the 1D convolutional backbone ensures fast inference.

In addition, we compare our method with the 11 baselines in training time in Table 5. As can be seen, DCdetector

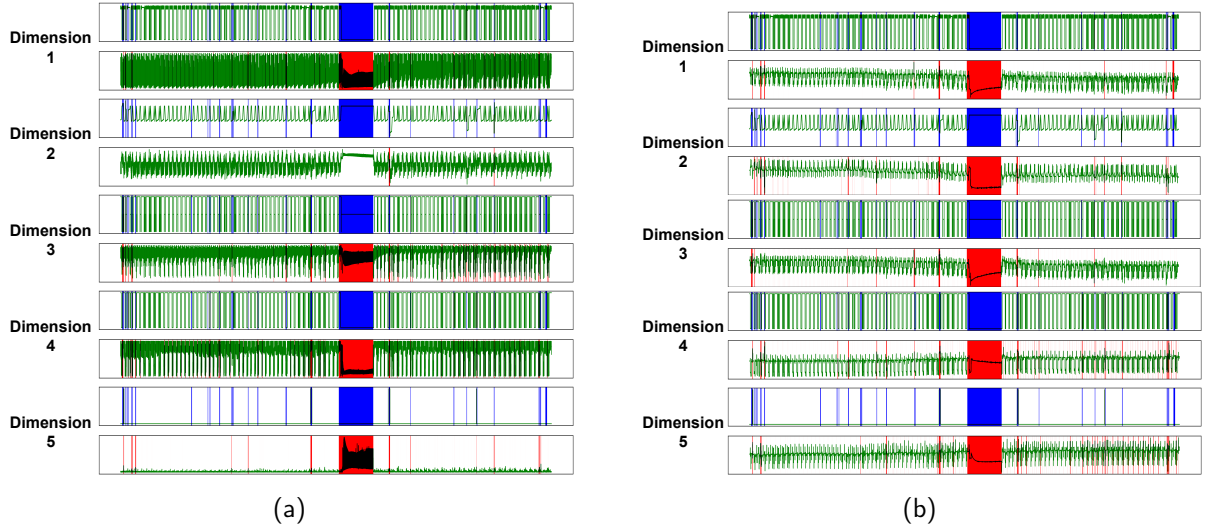


Figure 4: Comparison between the time series data in the SWaT_MV [63] data set processed by the TranAD [24] (a) and EPAAD-Net (b). Here, the red and blue blocks indicate the actual and predicted anomalies, respectively.

Table 4

Comparison between the EPAAD-Net and eleven deep learning baselines in number of parameters (kb). For each data set, the smallest number is highlighted in the **red, bold** font while the second smallest number is indicated in the *blue, italic* font.

Method	SMD	NAB	MBA	SMAP	SWaT_MV	SWaT_UV
OmniAnomaly	72.23	53.58	54.09	65.68	78.78	53.58
MSCRED	4,833.50	4,833.50	4,833.50	4,833.50	4,833.50	4,833.50
MAD-GAN	38.68	3.27	4.23	26.24	51.13	3.27
USAD	41.44	5.31	6.29	28.75	54.14	5.31
MTAD-GAT	99,715.86	0.14	<i>1.20</i>	18,879.98	321,864.79	0.14
CAE-M	28.24	28.24	28.24	28.24	28.24	28.24
GDN	16.70	2.24	2.63	11.62	21.77	2.24
TranAD	498.20	1.44	3.34	223.72	880.93	1.44
DTAAD	182.34	0.67	1.36	81.95	322.35	0.67
TimesNet	<i>21.29</i>	18.89	18.95	<i>20.45</i>	<i>19.21</i>	18.89
DCdetector	142.78	0.61	2.11	126.46	17.02	0.61
EPAAD-Net (Ours)	123.01	<i>0.32</i>	0.71	54.97	224.31	<i>0.32</i>

achieves the shortest training time, while EPAAD-Net ranks second on most data sets, consistently outperforming all other baselines. It is suggested that the complexity of our model is reasonable.

5.2.2. Inference Latency

According to the inference latency shown in Table 6, EPAAD-Net maintains low inference latency across all six data sets. Some baselines are marginally faster, but they compromise detection accuracy. Our EPAAD-Net achieves a favorable balance between latency and accuracy, which makes it suitable for edge applications.

We also measured inference latency across different window sizes $L \in \{10, 20, 50, 100, 200, 500\}$ on the six data sets, to evaluate the scalability of EPAAD-Net under varying input sequence lengths. As shown in Table 7, the latency remains nearly constant for low-dimensional data sets (i.e., SWaT_MV, MBA and NAB), with growth rates below 0.5%

when T increases from 10 to 500. For high-dimensional data sets, such as SWaT_UV, latency increases from 0.040 ms to 0.158 ms, consistent with the linear $O(n)$ complexity of the 1D convolutional backbone.

To provide a comprehensive efficiency evaluation, we further report the values of FLOPs, inference latency, throughput, and peak GPU memory consumption on different data sets in Table 8. It is shown that EPAAD-Net achieves consistently low inference latency and high throughput across all data sets, with FLOPs ranging from 0.02M to 1.20M depending on the feature dimension.

5.2.3. Inference Latency on Edge Device

To verify the performance of the proposed EPAAD-Net on the edge device, we also tested our model, together with 11 baselines, using an NVIDIA Jetson Xavier NX Toolkit. This toolkit contains an operating system of Ubuntu 18.04 LTS, the memory of 7.6GiB, two processors of ARMv8

Table 5

Comparison between EPAAD-Net and 11 baselines in training time (Seconds/Epoch). For each data set, the shortest time is indicated in the **red, bold** font while the second shortest time is highlighted in the *blue, italic* font.

Method	SMD	NAB	MBA	SMAP	SWaT_MV	SWaT_UV
OmniAnomaly	235.29	33.51	64.14	23.68	3,981.71	25.15
MSCRED	1,415.25	47.28	92.00	60.69	45,611.29	35.09
MAD-GAN	268.22	40.97	69.09	29.27	4,923.69	31.14
USAD	234.81	35.18	63.35	23.24	4,169.13	25.56
MTAD-GAT	4,038.52	42.81	66.53	71.46	208,177.20	31.34
CAE-M	190.87	17.29	35.18	13.29	5,282.12	12.79
GDN	68.84	6.63	42.11	5.75	1,356.54	4.91
TranAD	6.54	0.69	1.27	0.58	141.80	0.53
DTAAD	5.80	0.68	1.28	0.53	86.04	0.52
TimesNet	17.35	1.86	2.59	1.56	<i>1.56</i>	1.55
DCdetector	1.36	0.14	0.15	0.23	0.13	0.16
EPAAD-Net (Ours)	<i>4.63</i>	<i>0.51</i>	<i>1.06</i>	<i>0.49</i>	78.32	<i>0.43</i>

Table 6

Comparison between the EPAAD-Net and eleven deep learning baselines in inference latency (i.e., average testing time, ms) per record. For each data set, the shortest time is indicated in the **red, bold** font while the second shortest time is highlighted in the *blue, italic* font.

Method	SMD	NAB	MBA	SMAP	SWaT_MV	SWaT_UV
OmniAnomaly	0.19	0.19	0.19	0.19	0.19	0.19
MSCRED	12.27	1.28	1.29	4.17	1.32	1.28
MAD-GAN	0.11	0.07	0.07	0.07	0.07	<i>0.07</i>
USAD	<i>0.13</i>	<i>0.12</i>	<i>0.12</i>	<i>0.12</i>	<i>0.12</i>	0.12
MTAD-GAT	0.92	0.73	0.79	0.81	0.79	0.74
CAE-M	0.38	0.12	0.12	0.15	0.14	0.12
GDN	0.39	0.34	0.39	0.39	0.39	0.34
TranAD	0.91	0.91	0.91	0.92	0.91	0.92
DTAAD	0.72	0.73	0.73	0.73	0.74	0.74
TimesNet	4.41	4.39	4.41	4.45	4.44	4.42
DCdetector	0.77	0.78	0.77	0.77	0.78	0.78
EPAAD-Net (Ours)	0.38	0.06	0.21	0.14	0.19	0.04

*: The method consumed excessive computational resources and failed to produce a result under the experimental setup.

Table 7

Inference latency (ms) of the EPAAD-Net under varying window sizes L on six data sets.

Data Set	L					
	10	20	50	100	200	500
SMD	0.380	0.380	0.384	0.376	0.526	1.132
NAB	0.060	0.060	0.060	0.060	0.060	0.060
MBA	0.210	0.213	0.213	0.213	0.213	0.215
SMAP	0.140	0.140	0.142	0.140	0.142	0.271
SWaT_MV	0.190	0.192	0.190	0.192	0.194	0.192
SWaT_UV	0.040	0.041	0.041	0.042	0.067	0.158

Processor rev 0 (v8l) and a graphics card of NVIDIA Tegra Xavier (nvgpu). We used the six data sets described in Table 1. We computed the average time per record across three runs. The results are shown in Table 9. It is demonstrated that the EPAAD-Net is able to fulfill the real-time time series anomaly detection task on the edge device and our model runs faster than over a half of the baselines.

5.3. Test on the Real Industrial Time Series Data

We further tested the proposed method using the NVIDIA Jetson Xavier NX Toolkit on a real industrial time series data set. This data set was collected on a rubber tyre calendering production line in a factory located in Qingdao, China. The sensor suite encompasses pressure sensors, temperature sensors, current sensors, etc, which are standard instrumentation for monitoring process stability in tyre manufacturing. The dimensionality of the data is 36, comprising process variables and their derived features. The training set consists of 2,800 records collected under normal operating conditions, containing only normal data without anomalous samples, following the standard unsupervised anomaly detection protocol. The test set contains 2,800 records with an anomaly ratio of approximately 5.8%, where anomalies are defined as instances in which production parameters deviate from their normal operating ranges, leading to degradation of product quality. The results produced by the proposed method and two state-of-the-art baselines are shown in Table 10. It can be observed that our method achieves the better, or at

Table 8

The values of different efficiency metrics of EPAAD-Net, including FLOPs, inference latency, throughput, and peak GPU memory consumption, on different data sets.

Data Set	Dim. of Feat.	FLOPs (M)	Latency (ms)	Lat/Samp (μ s)	Throughput (Samp/s)	GPU Mem (MB)
SMD	38	0.58	0.38	2.97	336,700.34	62.21
NAB	1	0.01	0.06	0.47	2,127,659.57	9.55
MBA	2	0.01	0.21	1.64	609,756.10	10.93
SMAP	25	0.26	0.14	1.09	917,431.19	43.05
SWaT_MV	6	0.02	0.19	1.48	675,675.68	16.48
SWaT_UV	55	1.20	0.04	0.31	3,225,806.45	85.69

Table 9

The inference latency (ms) per record taken by EPAAD-Net and 11 baselines on an NVIDIA Jetson Xavier NX Toolkit. For each data set, the shortest time is indicated in the **red, bold** font while the second shortest time is highlighted in the *blue, italic* font.

Method	SMD	NAB	MBA	SMAP	SWaT_MV	SWaT_UV
OmniAnomaly	0.07	0.08	0.08	0.08	0.07	0.07
MSCRED	203.80	0.32	0.67	85.57	5.08	0.34
MAD-GAN	0.03	0.03	0.03	0.03	0.03	0.04
USAD	<i>0.05</i>	<i>0.05</i>	<i>0.05</i>	<i>0.05</i>	0.04	0.04
MTAD-GAT	8.70	4.27	4.21	4.47	4.36	3.71
CAE-M	6.29	0.09	0.06	2.86	0.24	<i>0.05</i>
GDN	2.29	2.06	2.01	2.02	2.03	1.87
TranAD	0.94	0.28	0.24	0.46	0.28	0.25
DTAAD	0.44	0.44	0.42	0.44	0.41	0.44
TimesNet	0.52	0.44	0.44	0.48	0.43	0.46
DCdetector	0.31	0.27	0.25	0.29	0.24	0.24
EPAAD-Net (Ours)	0.34	0.17	0.17	0.17	0.16	0.16

least competitive, results with fewer parameters and a faster inference speed, compared to its two counterparts.

It should be noted that the data set used in this experiment is relatively small compared to the public data sets that we used. This is due to the limited availability of labeled industrial data, as data collection on the production line requires coordination with factory operations and is also subject to confidentiality restrictions. Nevertheless, the real data set provides a valuable real-world validation of the proposed method under actual production conditions. The consistent improvement in performance over the baselines on this small-scale data set, together with the extensive results on the six public data sets, demonstrates the practical applicability of EPAAD-Net.

5.4. Ablation Study

We performed a series of ablation experiments to investigate the effect of different components of EPAAD-Net, the training stability of EPAAD-Net, the sensitivity of its hyperparameters, the robustness of EPAAD-Net to perturbations, and the cross-data-set generalizability of EPAAD-Net.

5.4.1. The Impact of Different Components of EPAAD-Net

To examine the impact of different components of the proposed EPAAD-Net, we derived eight variants of the EPAAD-Net by removing a component from this network or replacing a component by the other method. The description of each variant is shown in Table 11. The SED→Att variant

replaces the denoising module with a standard Transformer decoder layer. The number of attention heads is the largest integer h such that $d \bmod h = 0$. For simplicity, only the F1 score metric was used.

As displayed in Fig. 5 and detailed in Table 12, EPAAD-Net normally achieves the highest F1 scores across all six data sets compared to its eight variants. The performance of the most impactful components varies across data sets. In particular, SCTM is critical for SMD and SWaT_MV; the dual-branch structure (LTDMB and TSADB) plays a dominant role in NAB, MBA, SMAP, and SWaT_UV. In general, w/o SCTM, w/o LTDMB, and w/o TSADB produce the most significant performance degradation, confirming that the proposed SCTM and the dual-branch architecture are the key contributors to the effectiveness of EPAAD-Net.

A notable observation from Table 12 is that removing either the sine or cosine branch individually causes less degradation than removing the entire SCTM on some data sets (e.g., SMD, NAB and SWaT_UV). This is because the sine and cosine branches capture odd- and even-symmetric periodic components, respectively, and removing only one branch still preserves partial periodic information. However, removing the entire SCTM eliminates the periodic inductive bias completely.

The complexity and efficiency analysis of the eight variants in Table 13 shows that removing the dual-branch structure substantially reduces parameters and latency but causes significant accuracy degradation, while removing SCTM

Table 10

The results that two baselines and our method produced using a NVIDIA Jetson Xavier NX Toolkit on a real industrial time series data set. The best result is indicated in the **red, bold** font while the second best result is highlighted in the *blue, italic* font.

Method	F1-Score	Precision	Recall	AUC	Param (kb)	Speed (ms)
TranAD [24]	0.9267	0.9999	0.8635	<i>0.9317</i>	448.92	<i>1.21</i>
DTAAD [25]	<i>0.9301</i>	0.9839	<i>0.8818</i>	0.9084	<i>164.32</i>	1.58
EPAAD-Net (Ours)	0.9720	<i>0.9843</i>	0.9599	0.9454	114.02	0.33

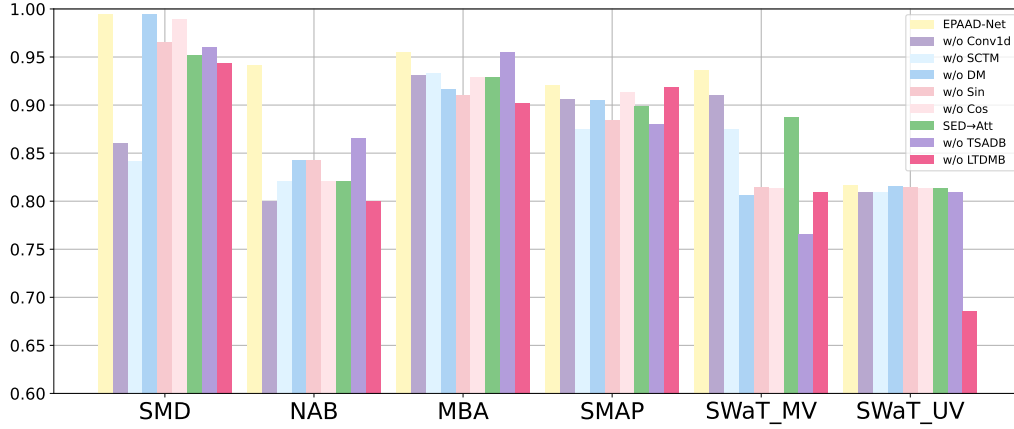


Figure 5: The comparison between the EPAAD-Net and its eight variants in terms of the F1 score values that they produced on different data sets.

Table 11

Eight variants of our EPAAD-Net obtained by removing or replacing a component of this network.

Variant	Description
w/o Conv1d	Removing the Conv1d blocks.
w/o SCTM	Removing the SCTM.
w/o DM	Removing the denoising module.
w/o Sin	Removing the Sine Transform block from the SCTM.
w/o Cos	Removing the Cosine Transform block from the SCTM.
SED → Att	Replacing the stacked encoder-decoder with the attention unit.
w/o TSADB	Removing the TSADB.
w/o LTDMB	Removing the LTDMB.

yields only marginal efficiency gains at the cost of noticeable F1 drops on periodic data sets, confirming that SCTM contributes accuracy benefits with negligible computational overhead. The SED→Att variant increases both number of parameters and latency yet underperforms the full model, justifying our choice of the encoder-decoder structure.

Although this experiment confirms the contributions of the SCTM and the dual-branch architecture, deployment of EPAAD-Net on the frontline of fog infrastructure introduces additional challenges, including adversarial vulnerabilities such as data poisoning and evasion attacks. Recent advances

in bio-inspired learning paradigms, exemplified by neuro-morphic computing with spiking neural networks and quantum noise injection, have demonstrated resilience against such threats while enabling ultra-fast, energy-efficient inference on specialized processors [74]. Federated machine learning frameworks have been proposed to address the scalability and privacy constraints of industrial intrusion detection, allowing distributed edge nodes to collaborate without exposing sensitive data [75]. Even if these aspects are not addressed in this study, we identify them as promising future directions, particularly as industrial edge networks evolve toward distributed and privacy-aware architectures.

5.4.2. Training Stability

To assess the statistical stability of the training process of EPAAD-Net, we conducted five independent training runs on all six data sets with different random seeds. The mean and standard deviation of the F1 score and AUC values computed across the five runs are reported in Table 14. The statistical results demonstrate that EPAAD-Net achieves consistently high performance across all six data sets, with small standard deviations in terms of both the F1 score and AUC metrics across five independent runs. The small standard deviations confirm that the training process of EPAAD-Net is stable and reproducible.

5.4.3. Sensitivity of Key Hyperparameters

To systematically evaluate the sensitivity of EPAAD-Net to key hyperparameters, we conducted an additional

Table 12

The F1 score values derived using eight variants of EPAAD-Net across six data sets. The best result for each data set is indicated in **red, bold** font while the second best result is highlighted in *blue, italic* font.

Data Set	EPAAD-Net	w/o Conv1d	w/o SCTM	w/o DM	w/o Sin	w/o Cos	SED→Att	w/o TSADB	w/o LTDMB
SMD	<i>0.9944</i>	0.8606	0.8410	0.9945	0.9656	0.9892	0.9520	0.9603	0.9437
NAB	0.9412	0.8002	0.8204	0.8420	0.8420	0.8204	0.8204	<i>0.8646</i>	0.8005
MBA	0.9548	0.9314	<i>0.9327</i>	0.9164	0.9104	0.9290	0.9290	0.9548	0.9018
SMAP	0.9200	0.8742	0.9051	0.8842	0.9131	0.8985	0.8799	0.9055	<i>0.9164</i>
SWaT_MV	0.9364	0.7653	0.8092	<i>0.8158</i>	0.8088	0.8085	0.8151	0.8141	0.8131
SWaT_UV	0.8162	0.8088	0.8138	0.8138	<i>0.8158</i>	0.8141	0.8151	0.6849	0.8138

Table 13

The comparison between EPAAD-Net and its eight variants in complexity and efficiency (averaged across all six data sets).

Variant	Avg. Params.	Avg. Latency (ms)
EPAAD-Net	17,492	0.380
w/o Conv1d	7,095	0.223
w/o SCTM	15,780	0.347
w/o DM	16,331	0.327
w/o Sin	16,636	0.360
w/o Cos	16,636	0.360
SED→Att	22,046	0.401
w/o TSADB	12,823	0.223
w/o LTDMB	7,095	0.223

Table 14

Statistical analysis of the training stability of EPAAD-Net over five independent runs with different random seeds.

Data Set	F1 (Mean ± Std)	AUC (Mean ± Std)
SMD	0.9737 ± 0.0228	0.9938 ± 0.0026
NAB	0.9043 ± 0.0168	0.9941 ± 0.0006
MBA	0.9533 ± 0.0033	0.9766 ± 0.0017
SMAP	0.9107 ± 0.0082	0.9882 ± 0.0010
SWaT_MV	0.9194 ± 0.0084	0.9522 ± 0.0017
SWaT_UV	0.8155 ± 0.0016	0.8479 ± 0.0048

experiment on the SMD data set by varying the sliding window size $L \in \{5, 10, 15, 20, 30, 50\}$ and the learning rate $\eta \in \{1 \times 10^{-6}, 1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}\}$. The results are reported in Table 15. As can be seen, the model is relatively robust to the variation of window sizes, with F1 score values exceeding 0.90 for all window sizes tested. However, the selection of learning rate is sensitive. The values below 5×10^{-5} lead to training failure, whereas the model performs stably when the value ranges from 5×10^{-5} to 1×10^{-3} with F1 score values exceeding 0.97. These results confirm that EPAAD-Net is generally robust to the two hyperparameters, except for requiring an appropriate learning rate.

5.4.4. Robustness to Perturbations

We further evaluated the robustness of our method to different perturbations, including distribution shifts (trend, scale, shift and seasonal), varying anomaly ratios (1%–50%), noisy measurements (SNR 5–30 dB), and missing observations (5%–30%), on the SMD data set. Table 16 summarizes

Table 15

Sensitivity analysis of key hyperparameters of EPAAD-Net on the SMD data set.

Window Size (L)			Learning Rate (η)		
L	F1	AUC	η	F1	AUC
5	0.9898	0.8900	1×10^{-6}	0.0000	0.3250
10	0.9944	0.9982	1×10^{-5}	0.0000	0.5613
15	0.9246	0.8846	5×10^{-5}	0.9922	0.8915
20	0.9121	0.8839	1×10^{-4}	0.9944	0.9982
30	0.9074	0.8837	5×10^{-4}	0.9870	0.9248
50	0.9038	0.8840	1×10^{-3}	0.9991	0.9314

Table 16

Evaluation of the robustness of EPAAD-Net to different perturbations on the SMD data set.

Perturbation	Range	Worst F1	Avg F1
Distribution Shift	Intensity 0.1–0.5	0.9254	0.9371
Anomaly Ratio	1%–50%	0.5476	0.8602
Noisy Measurement	SNR 5–30 dB	0.6837	0.8631
Missing Observation	5%–30%	0.6131	0.7377

the F1 score values obtained in these conditions. It can be seen that EPAAD-Net remains robust under moderate perturbations but degrades under extreme missing data (30%) and low anomaly ratios (1%).

5.4.5. Cross-Data-Set Generalizability

To evaluate the cross-data-set generalizability of our method under varying feature dimensionalities, we trained EPAAD-Net on the full SMD data set (38 dimensions) and tested the model on the other data sets, truncating the source features to match each target dimension. As shown in Table 17, EPAAD-Net achieves strong cross-data-set transfer from SMD to SWaT_UV with an F1 score of 0.7649 and to MBA with an F1 score of 0.8762, but degrades significantly from SMD to SMAP and SWaT_MV with F1 score values of 0.1617 and 0.2436, respectively. It is indicated that cross-domain generalization is highly dependent on the similarity between the distributions of the source and target domains.

Table 17

Cross-data-set generalization results obtained when EPAAD-Net was trained on SMD and tested on other data sets.

Transfer Direction	F1 (PA)	AUC
SMD → NAB	0.6400	0.5386
SMD → MBA	0.8762	0.4498
SMD → SMAP	0.1617	0.5689
SMD → SWaT_MV	0.2436	0.5421
SMD → SWaT_UV	0.7649	0.8177
SMD → SMD	0.9944	0.9982

6. Conclusion

In this paper, we introduced a new anomaly detection neural network, referred to as Efficient Periodicity-Aware Anomaly Detection Network, or EPAAD-Net for short, to satisfy the demand for efficient and effective time series anomaly detection at the edge of the production line. In particular, we proposed a Sine-Cosine Transform Module (SCTM), which introduces a periodic inductive bias by projecting the features extracted onto the sine and cosine basis functions, enabling the network to better capture periodic patterns in time-series data for anomaly detection. Furthermore, we inserted a cross-attention module into the EPAAD-Net for the sake of capturing the long-term dependencies between data points.

The EPAAD-Net was compared against 14 baselines on six publicly available data sets. The results showed that EPAAD-Net achieves competitive detection accuracy compared to its counterparts across these data sets, with particular advantages in inference efficiency. It should be noted that EPAAD-Net outperformed the strongest baseline by 1.82% in F1 score on the challenging SWaT_MV data set. In particular, EPAAD-Net manifested the higher efficiency than the baselines, which is useful for deploying the anomaly detection network on the edge of the production line. The promising results should be due to both the simple network architecture and the proposed periodicity-aware SCTM.

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